

```
1 -----
2 -- 1-bit Full Adder/Subtractor
3 --
4 -- This entity describes a 1-bit full adder/subtractor.
5
6 -- Revision History:
7 --   Date:           Author:      Description:
8 --   07 Jan 2024    Chris M.      Initial revision; described entity and processes.
9 -----
10 library ieee;
11 use ieee.std_logic_1164.all;
12 use ieee.numeric_std.all;
13
14 -- AdderSubtractor entity declaration.
15 --
16 -- This entity describes a 1-bit adder subtracter.
17 --
18 -- Inputs:
19 --   A : 1-bit A input
20 --   B : 1-bit B input
21 --   Cin : 1-bit Carry input
22 --   AddBarSub : operation select signal. Add if 0, sub if 1.
23
24 -- Outputs:
25 --   Cout      : 1-bit carry/borrowbar out
26 --   S         : 1-bit result
27 --
28 entity AdderSubtractor is
29   port(
30     A      : in  std_logic;
31     B      : in  std_logic;
32     Cin     : in  std_logic;
33     AddBarSub : in  std_logic;
34     Cout    : out std_logic;
35     S       : out std_logic
36   );
37 end AdderSubtractor;
38
39 architecture concurrent of AdderSubtractor is
40 begin
41   S <= A xor B xor Cin xor AddBarSub;
42   Cout <= (A and Cin) or ((B xor AddBarSub) and (A or Cin));
43 end concurrent;
44
```